

CareerCompass: Student Aptitude & Reasoning Assessment

1. Numerical Trend Recognition (Aptitude: Numerical Reasoning)

Question: A student club's social media followers grow as follows: 3, 9, 21, 45, 93. What is the target for the next month based on this logical progression?

- **Options:**
 - A) 189
 - B) 141
 - C) 195
 - D) 120
- **Aptitude Reasoning:** Tests **Numerical Reasoning**. The pattern is . .
- **Psychological Mapping:** Investigative (I) / Thinker.
- **Career Domain:** Analytical & Technical (Data Scientist, Actuary).
- **Weighting:** A: 1.0, B: 0.2, C: 0.3, D: 0.1.
- **Hidden Insight:** Distinguishes between "ambitious guessing" and "calculated logic".
- **Confidence Impact:** 0.95.
- **Anti-Manipulation:** Mathematical truth cannot be faked through personality bias.

2. The Missing Section (Aptitude: Verbal Reasoning)

Question: A teacher's feedback on your report says: "Your arguments are technically sound, but the final conclusion feels somewhat detached from your primary evidence." What is the teacher's core suggestion?

- **Options:**
 - A) Improve your grammar and spelling.
 - B) Strengthen the logical link between your facts and your summary.
 - C) Find more evidence to support a new topic.
 - D) The topic is too complex for the evidence provided.
- **Aptitude Reasoning:** Tests **Verbal Reasoning** (VR) and nuance inference.
- **Psychological Mapping:** Investigative (I) / Social (S). Evaluates **Scanning** for relevant cues.

- **Career Domain:** Social & Psychological / Leadership (Lawyer, Researcher).
- **Weighting:** B: 1.0, A: 0.1, C: 0.4, D: 0.2.
- **Hidden Insight:** Detects **Metacognitive Skills**—the ability to monitor one's own work for logical gaps.
- **Confidence Impact:** 0.90.
- **Anti-Manipulation:** Requires accurate text interpretation rather than "polite" answering.

3. Spatial Event Planning (Aptitude: Space Relations)

Question: You must set up 10 booths in a school gym. Two large pillars block the center. If you mentally rotate the booths to create a circular flow around the pillars, how many 2x2 meter booths can fit in a 10x10 meter area while maintaining a 2-meter walking path throughout?

- **Options:**
 - A) 12
 - B) 10
 - C) 8
 - D) 6
- **Aptitude Reasoning:** Tests **Space Relations** (SR)—mentally manipulating objects in 3D space.
- **Psychological Mapping:** Realistic (R) / Doer.
- **Career Domain:** Analytical & Technical (Architect, Engineer, Web Developer).
- **Weighting:** B: 1.0, A: 0.4, C: 0.6, D: 0.3.
- **Hidden Insight:** Correlates spatial ability with **Problem-Solving Speed** in physical systems.
- **Confidence Impact:** 0.95.

4. The Group Conflict (Hybrid: Social + Leadership)

Question: During a rehearsal for a group presentation, a teammate makes a sarcastic joke about your section of the work. The rest laugh. You feel this undermines your role as leader. How do you respond?

- **Options:**

- A) Wait until the rehearsal finishes, then speak to the teammate privately about the team's focus.
- B) Make a joke back at yourself to show the team you are easygoing.
- C) Immediately stop the rehearsal to remind everyone of the grading criteria.
- D) Ignore the comment and continue with increased focus.
- **Aptitude Reasoning:** Requires **Social Forecasting**—predicting the fallout of authority vs. harmony.
- **Psychological Mapping:** Enterprising (E) + Social (S). Measures **Social Regulation**.
- **Career Domain:** Leadership & Business / Social (Product Manager, Teacher).
- **Weighting:** A: 1.0, B: 0.6, C: 0.4, D: 0.3.
- **Hidden Insight:** Identifies if the leader prioritizes **Task vs. Relationship** under stress.
- **Anti-Manipulation:** Option B is a "Social Desirability Trap" that identifies those who prioritize being "well-liked" over being effective.

5. Abstract Pattern Sequence (Aptitude: Abstract Reasoning)

Question: Sequence: Circle inside Square Square inside Triangle Triangle inside Hexagon. What completes the series?

- **Options:**
 - A) Hexagon inside Circle
 - B) Hexagon inside Octagon
 - C) Octagon inside Square
 - D) Square inside Circle
- **Aptitude Reasoning:** Tests **Abstract Reasoning** (AR). Pattern: Outer shape becomes inner; new outer shape adds +1/+2 sides.
- **Psychological Mapping:** Investigative (I) / Thinker.
- **Career Domain:** Analytical & Technical (Software Engineer, AI Engineer).
- **Weighting:** B: 1.0, A: 0.2, C: 0.4, D: 0.3.
- **Hidden Insight:** AR is a strong predictor of success in non-verbal problem-solving.

- **Confidence Impact:** 0.95.

6. Mechanical Advantage (Aptitude: Mechanical Reasoning)

Question: To lift a 50kg crate using a long wooden plank as a lever, where should the pivot point (fulcrum) be placed to use the **least** amount of physical force?

- **Options:**
 - A) Closest to your hands.
 - B) Exactly in the middle.
 - C) Closest to the heavy crate.
 - D) Force remains constant regardless of placement.
- **Aptitude Reasoning:** Tests **Mechanical Reasoning** (MR) and physical principles.
- **Psychological Mapping:** Realistic (R) / Doer.
- **Career Domain:** Analytical & Technical (Surgeon, Mechanical Engineer, Carpenter).
- **Weighting:** C: 1.0, A: 0.2, B: 0.5, D: 0.0.
- **Hidden Insight:** Predicts success in vocational tasks involving physical manipulation.
- **Confidence Impact:** 0.95.

7. Data Interpretation (Aptitude: Analytical Reasoning)

Question: Club attendance Month 1: 20, Month 2: 30, Month 3: 15. In Month 3, a mandatory exam session was held the same day. A teammate says, "The club is dying." What is your logical conclusion?

- **Options:**
 - A) The teammate is right; interest is declining.
 - B) The exam is a "confounding variable"; we must test Month 4 without exams to know the trend.
 - C) We should change the club topic to something more exciting.
 - D) The growth in Month 2 was a fluke.
- **Aptitude Reasoning:** Tests **Critical Thinking** and identifying variables.

- **Psychological Mapping:** Investigative (I) / Thinker. Evaluates **Appraising** incoming data.
- **Career Domain:** Analytical & Technical / Business (Data Scientist, Business Analyst).
- **Weighting:** B: 1.0, A: 0.2, C: 0.3, D: 0.4.
- **Hidden Insight:** Measures the "Scanning Phase"—distinguishing between relevant and irrelevant situational cues.
- **Confidence Impact:** 0.90.

8. The Design Dilemma (Hybrid: Creative + Analytical)

Question: You are designing a logo for a local charity. They want "traditional," but your research shows a "modern" logo will attract 40% more young donors.

- **Options:**
 - A) Create a hybrid logo using a classic symbol in a clean, modern style.
 - B) Follow the instructions exactly to ensure the client is happy.
 - C) Use the modern style only, as you are the expert.
 - D) Create two different logos and let them choose without giving your opinion.
- **Aptitude Reasoning:** Uses **Abductive Reasoning** to find the "likeliest explanation/solution" for conflicting data.
- **Psychological Mapping:** Artistic (A) + Investigative (I).
- **Career Domain:** Creative & Artistic / Business (UI/UX Designer, Marketing Strategist).
- **Weighting:** A: 1.0, B: 0.6, C: 0.4, D: 0.5.
- **Hidden Insight:** Measures **Elaborated Knowledge Structures**—the ability to integrate conflicting information into a new solution.

9. Clerical Accuracy Check (Aptitude: PSA)

Question: Compare these two student lists of IDs: List 1: 4492-X, 3921-A, 8820-Z, 1102-B. List 2: 4492-X, 3921-A, 8802-Z, 1102-B. How many errors are there in List 2?

- **Options:**
 - A) 0
 - B) 1

- C) 2
- D) 3
- **Aptitude Reasoning:** Tests **Perceptual (Clerical) Speed and Accuracy** (PSA).
- **Psychological Mapping:** Conventional (C) / Organizer.
- **Career Domain:** Leadership & Business / Technical (Accounting, Data Entry, HR Specialist).
- **Weighting:** B: 1.0, A: 0.1, C: 0.2, D: 0.1.
- **Hidden Insight:** High scores predict success in detail-oriented, systematic environments.
- **Confidence Impact:** 0.95.

10. The Ethical Slide (Psychology: Integrity + Strategic)

Question: You realize a teammate accidentally inflated your group's project success in a report to the teacher. Your teacher has already given the group an 'A'. What do you do?

- **Options:**
 - A) Inform the teacher privately and offer to fix the data, even if the grade changes.
 - B) Say nothing; it was an accident and the outcome was good.
 - C) Correct the teammate in private but leave the grade as is.
 - D) Hope the teacher doesn't notice and work extra hard on the next project to "earn" the grade.
- **Aptitude Reasoning:** Tests **Critical Thinking** regarding long-term integrity risk.
- **Psychological Mapping:** Conventional (C) + Social (S). Evaluates **Perspective-Taking**.
- **Career Domain:** Leadership & Business (HR, Auditor).
- **Weighting:** A: 1.0, B: 0.1, C: 0.5, D: 0.3.
- **Hidden Insight:** High **Conscientiousness** is a core predictor of job stability.

11. Logical Syllogism (Aptitude: Deductive Reasoning)

Question: All successful artists are creative. Some students are successful artists. Therefore, some students are creative. Is this:

- **Options:**

- A) Always True
- B) Always False
- C) Uncertain
- **Aptitude Reasoning:** Tests **Deductive Reasoning** and categorical logic.
- **Psychological Mapping:** Investigative (I).
- **Career Domain:** Analytical & Technical / Social (Psychologist, Researcher).
- **Weighting:** A: 1.0, B: 0.0, C: 0.2.
- **Hidden Insight:** Logic checks identify resistance to "optimism bias".

12. The Feedback Loop (Hybrid: Social + Analytical)

Question: You worked with Ashley on a project. She was frustrated and quiet. If her anger is due to the difficult math tasks, you can help. What is your **first** step?

- **Options:**
 - A) Ask Ashley if she needs help with the project tasks.
 - B) Let her know you perceive her frustration and ask if she wants to talk.
 - C) Finish the project tasks yourself to reduce her stress.
 - D) Tell the teacher she is being difficult and ask for a new partner.
- **Aptitude Reasoning:** Requires **Social Data-Gathering** before action.
- **Psychological Mapping:** Social (S). Evaluates the **Scanning** phase.
- **Career Domain:** Social & Psychological (Counselor, HR Specialist, Teacher).
- **Weighting:** B: 1.0, A: 0.7, C: 0.4, D: 0.1.
- **Hidden Insight:** Measures **Perspective-Taking**—accurately understanding another's internal frame of reference.

13. Language Consistency (Aptitude: Language Usage)

Question: Which sentence is most professional for a student scholarship application?

- **Options:**
 - A) I want this because your the best school in the city.
 - B) My objective is to utilize my competencies in a way that provides value to your organization.
 - C) I can do the work better then any other student in my class.

- D) I'm normally able to "get into someone's shoes" and experience their emotions.
- **Aptitude Reasoning:** Tests **Language Usage** (LU) and professional norms.
- **Psychological Mapping:** Conventional (C) / Organizer.
- **Career Domain:** Leadership & Business (PR, Administrator).
- **Weighting:** B: 1.0, A: 0.0, C: 0.1, D: 0.6.
- **Hidden Insight:** Identifies **Attention to Detail** and awareness of formal social standards.

14. Abstract Reasoning: Matrix Completion (Aptitude: AR)

Question: (Visual: A grid of 3x3 where the shapes rotate 90 degrees clockwise in each step). Which shape fits the final square?

- **Options:** (Descriptions of rotations)
- **Aptitude Reasoning:** Tests **Abstract Reasoning** (AR)—finding relations in non-verbal stimuli.
- **Psychological Mapping:** Investigative (I).
- **Career Domain:** Analytical & Technical (Software Engineer, AI Engineer).
- **Weighting:** 1.0 for the correct rotation.
- **Confidence Impact:** 0.95.

15. The "Expert" Trap (Hybrid: Social + Strategic)

Question: You know the correct answer to a lab problem, but your group leader is explaining it incorrectly. How do you respond?

- **Options:**
 - A) Politely ask a question that highlights the error, allowing the leader to discover the correction themselves.
 - B) State the correct answer out loud immediately so the group doesn't waste time.
 - C) Say nothing; it's better to follow the leader even if they are wrong.
 - D) Correct the leader and then report them to the teacher for poor management.
- **Aptitude Reasoning:** Requires **Metaperceptual Skills**—awareness of how others perceive you when you correct them.

- **Psychological Mapping:** Social (S) + Enterprising (E).
- **Career Domain:** Leadership & Business / Social (Product Manager, Art Therapist).
- **Weighting:** A: 1.0, B: 0.6, C: 0.4, D: 0.2.
- **Hidden Insight:** Evaluates **Social Regulation**—modifying behavior to promote a desired outcome.

16. Mechanical Gears (Aptitude: MR)

Question: If the first gear in a system of four connected gears rotates clockwise, which direction does the fourth gear rotate?

- **Options:**
 - A) Clockwise
 - B) Counter-clockwise
 - C) It doesn't move
 - D) It alternates
- **Aptitude Reasoning:** Tests **Mechanical Reasoning** (MR) and systemic logic. (Answer: B, each gear reverses direction).
- **Psychological Mapping:** Realistic (R) / Doer.
- **Career Domain:** Analytical & Technical (Engineer, Surgeon).
- **Weighting:** B: 1.0.
- **Confidence Impact:** 0.95.

17. Strategic Risk (Hybrid: Leadership + Strategic)

Question: Your club has 3 months of money left. You can do a "Safe" small sale with guaranteed \$50 profit or an "Ambitious" event with \$500 potential profit but a 50% risk of losing all current funds.

- **Options:**
 - A) Survey members to see what would make them attend the Ambitious event to reduce risk.
 - B) Do the Ambitious event; high risk is the only way to save the club.
 - C) Do the Safe sale to ensure the club stays open as long as possible.
 - D) Close the club now and save the money for next year.

- **Aptitude Reasoning:** Requires **Social Forecasting** and data-based decision making.
- **Psychological Mapping:** Enterprising (E) + Investigative (I).
- **Career Domain:** Leadership & Business (Entrepreneur, Marketing Strategist).
- **Weighting:** A: 1.0, B: 0.8, C: 0.5, D: 0.2.
- **Hidden Insight:** Distinguishes between **Risk-Averse** and **Strategic-Innovator** profiles.

18. The New Student (Psychology: Cultural Intelligence)

Question: A new student from a different culture joins your group. They are quiet and don't make eye contact. You:

- **Options:**
 - A) Research their culture's communication norms before judging their participation.
 - B) Speak louder and more slowly to ensure they understand you.
 - C) Assume they are shy and do the talking for them.
 - D) Tell them eye contact is necessary for leadership roles.
- **Aptitude Reasoning:** Requires **Perspective-Taking**.
- **Psychological Mapping:** Social (S) + Openness.
- **Career Domain:** Social & Psychological (Psychologist, Teacher, Counselor).
- **Weighting:** A: 1.0, B: 0.3, C: 0.4, D: 0.1.
- **Hidden Insight:** Detects the **Scanning Phase**—ability to distinguish cultural cues from personality traits.

19. Arithmetic Reasoning (Aptitude: Numerical Reasoning)

Question: A student store buys pens for \$0.50 and sells them for \$1.20. If they sell 40 pens but 5 pens are stolen, what is their total profit?

- **Options:**
 - A) \$28.00
 - B) \$25.50
 - C) \$23.00
 - D) \$30.00

- **Aptitude Reasoning:** Tests **Numerical Reasoning** and **Attention to Detail**.
(Total sales: $40 * 1.20 = 48$. Total cost: $45 * 0.50 = 22.50$. Profit = \$25.50).
- **Weighting:** B: 1.0.
- **Confidence Impact:** 0.95.

20. The Leadership Choice (Hybrid: Leadership + Social)

Question: You must pick 3 people for your debate team. Do you pick:

- **Options:**
 - A) Your 3 best friends so you can work together easily.
 - B) The 3 students with the highest history grades.
 - C) One great speaker, one great researcher, and one person who is good at organizing notes.
 - D) People who haven't had a chance to be on a team yet to be fair.
- **Aptitude Reasoning:** Tests **Ability to Identify Criteria** (ATIC)—recognizing task requirements.
- **Psychological Mapping:** Enterprising (E) + Conventional (C).
- **Career Domain:** Leadership & Business (Product Manager, Project Lead).
- **Weighting:** C: 1.0, B: 0.7, D: 0.6, A: 0.3.
- **Hidden Insight:** Measures if the student "Scans" for **Competency vs. Comfort**.

Advanced System Logic Notes

- **Career Mapping:** Students with high **AR + NR** scores map to **AI/Data Science**. High **VR + SR** scores map to **Architecture/Design**. High **CST (Social Reasoning) + VR** map to **Psychology/Law**.